

SmartBranch 360

Secure Branch Network — Project Presentation

PRESENTED BY

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PROJECT INTRODUCTION

SmartBranch 360 is a secure branch-office network built to give wired employees, guest Wi-Fi users, an internal server, and network administrators the right level of access — with internet reachability via NAT, DHCP-assigned addressing, and security controls that isolate guest traffic and restrict device management to authorized staff.

This project combines a Packet Tracer network build with an independent Python assurance tool that validates the configuration against the design.

The Problem Statement

What SmartBranch 360 was required to deliver

1

Wired employee access

At least 8 endpoints across employee, guest, and server roles.

2

Guest Wi-Fi, isolated

Guest devices reach the internet only — never the server or management network.

3

Internal server + NAT

One internal server, plus NAT overload for shared internet access.

4

DHCP-assigned addressing

End-user devices receive addresses automatically per VLAN.

5

Secure device management

SSH to network devices restricted to an authorized management subnet.

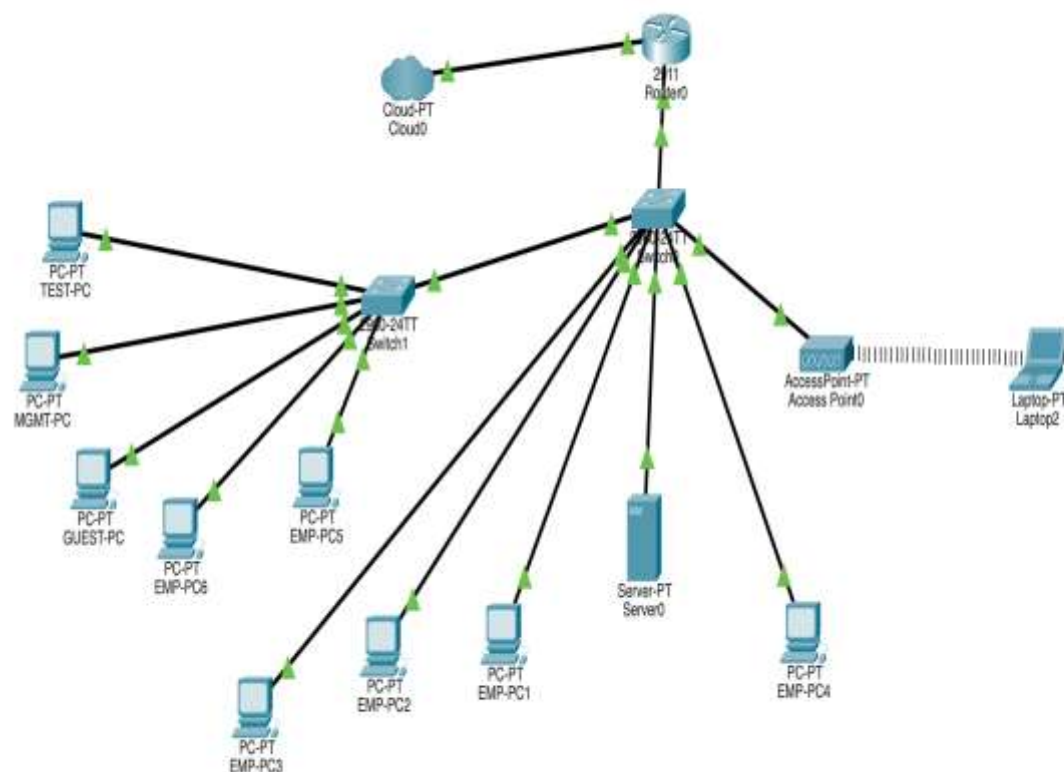
6

Automated verification

A Python tool to validate configuration, plus 5+ documented fault scenarios.

Network Design

Router-on-a-stick topology with four VLANs



VLAN	Name	Subnet
10	Employee	10.10.10.0/24
20	Guest	10.10.20.0/24
30	Server	10.10.30.0/24
99	Management	10.10.99.0/24

Security

- ◆ Guest → internet only; blocked from Server (30) and Management (99)
- ◆ Server reachable only from Employee (10) and Management (99)
- ◆ SSH to devices permitted only from the Management subnet

Router-on-a-stick + NAT overload + inter-VLAN ACLs

Python Assurance Tool

network_checker.py — validates config against the design

What it checks

- ✓ Plan consistency — valid subnets, gateways, no overlaps
- ✓ Trunk configuration — VLANs missing from allowed lists
- ✓ DHCP health — missing or exhausted address pools
- ✓ ACL correctness — rules that would block DNS (port 53)
- ✓ NAT status — missing or inactive overload/PAT
- ✓ Management access — SSH restricted to the Mgmt subnet

Sample finding

[FAIL] VLAN 20 missing on trunk SW1-SW2

Symptom: guest traffic silently dropped

Fix: `switchport trunk allowed vlan add 20`

Result

After the fix was applied in Packet Tracer, re-running the tool reported 0 FAIL and 0 WARN findings — confirming the fault was real and the fix fully resolved it. The tool exits non-zero on any FAIL, so it doubles as a quick pass/fail gate.

Fault Injection & Testing

Five faults intentionally introduced, diagnosed, and fixed

#	Fault	Caught by checker?
1	Wrong default gateway on an employee PC	Manual check (outside checker's scope)
2	Missing VLAN on the SW1–SW2 trunk	Yes — FAIL, trunk check
3	Bad / exhausted DHCP pool	Yes — FAIL/WARN, DHCP check
4	ACL misconfiguration blocking DNS	Yes — FAIL, ACL check
5	NAT misconfiguration (missing overload)	Yes — FAIL, NAT check



Each fault was resolved and re-verified with ping (and, where applicable, a clean checker run) before moving to the next scenario — with a known-good Packet Tracer save kept as a rollback point throughout.

Deliverables & Conclusion

- ✓ SmartBranch360.pkt — Packet Tracer build
- ✓ Packet Tracer Build Guide + CLI configs
- ✓ network_checker.py — Python validation tool
- ✓ Fault Cards — 5 scenarios with diagnosis and fix
- ✓ Design Document — topology, VLAN/IP plan, security rules
- ✓ requirements.yaml — network requirement/plan file
- ✓ Sample show-output files + sample validation report

Conclusion

SmartBranch 360 demonstrates a complete secure branch network: correct VLAN segmentation and routing, working NAT and DHCP, enforced boundaries between guest, employee, server, and management traffic — and an automated Python tool that verifies the configuration matches the design, catching real, representative faults rather than only the cases it was written around.

*Thank
you!*